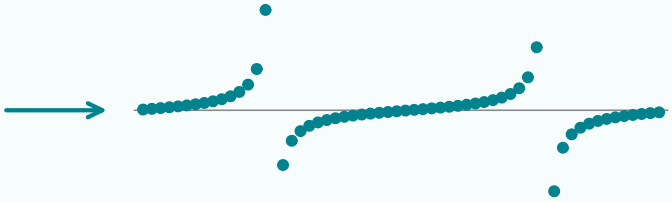
Chapter 23

Tables and charts

be a graph detective

0.10	0.20	0.31	0.42	0.55	0.68
0.84	1.03	1.26	1.56	1.96	2.57
3.60	5.80	14.10	-34.23	-7.70	-4.29
-2.93	-2.19	-1.71	-1.37	-1.12	-0.92
-0.75	-0.60	-0.47	-0.36	-0.25	-0.14



the same 60 numbers as dots: a repeating pattern of spikes

A list of four numbers you can hold in your head. Lecture 5 points out that research says we can only juggle about **four to seven things** at once, so a list of sixty numbers is just noise. Turn the same numbers into dots (above) and your eyes spot the pattern instantly.

This chapter is about **drawing data honestly** and **reading other people's graphs suspiciously**. Each page is a **case file**: a question, the evidence, and the verdict.

Goes with **Lecture 5** (graphical analysis: basic charts, outliers, bad and good graphs, form / message / criticism).

Two kinds of data

Data just means the facts you've recorded. Before drawing anything, ask one question: **is each fact a word or a number?**

Categorical (words / labels)

favourite drink: tea, coffee, juice

bus route: 7, 36, 58 (numbers used as names!)

eye colour, course, month

Numerical (measured / counted)

height 1.74 m, age 19, wage £16,400

you can sensibly add or average them

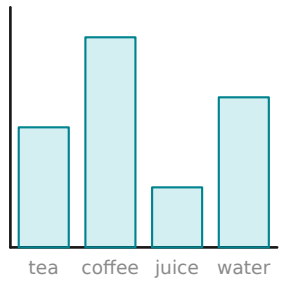
(the average bus route is nonsense)

A **frequency** is a count: "how many times did this happen?". Most charts draw frequencies. Lecture 5 asks the difference between a **chart** and a **graph**: loosely, a **graph** plots numbers against numbers on two axes, while **chart** is the wider word (bar charts, pie charts, tables...). People use the words loosely, so don't lose sleep over it.

The line-up: six basic charts

The same small data set drawn six ways (the lecture's Minitab examples). Each chart is good at **one** job.

Bar chart



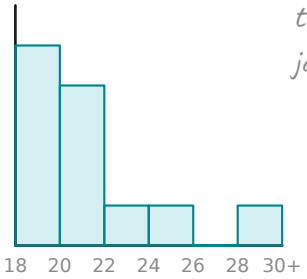
categories; gaps between bars because tea and coffee are separate things

Pie chart



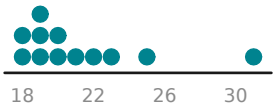
parts of one whole; slice angle = share of 360°

Histogram



numbers in ranges; bars touch because the ranges join up

Dotplot



one dot per value; small data sets, see clusters

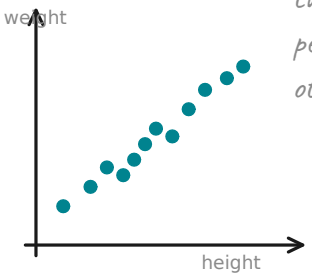
Stem and leaf

like a dotplot made of the digits themselves

1	8 8 9 9 9
2	0 0 1 2 3 5
3	1

key: 2 | 5 means 25

Scatter plot



two measurements per person: does one go with the other?

Pie angle: coffee is 7 of 18 drinks, so its slice is $\frac{7}{18} \times 360° = 140°$.

Pie charts: the angle rule, both ways

RULE: pie chart angles

The whole circle, $360°$, stands for the **total** frequency.

Frequency to angle: $\text{angle} = \dfrac{\text{frequency}}{\text{total}} \times 360°$.

Angle to frequency: $\text{frequency} = \dfrac{\text{angle}}{360°} \times \text{total}$.

The angles add up to $360°$ and the frequencies add up to the total, so one missing slice is whatever is left over.

Example: a pie chart shows 250 people. One slice is $72°$. How many people is that?

$$\frac{72}{360} \times 250 = \frac{1}{5} \times 250 = 50$$ ← angle ÷ 360 = the fraction of the whole

Example: a $45°$ slice stands for 30 people. How many people does the whole pie show?

$$\frac{45}{360} = \frac{1}{8} \Rightarrow \text{total} = 30 \times 8 = 240$$ ← the slice is 1/8 of the circle

Example: slices of $150°$ and $120°$ and one more slice, for 120 people. The last slice?

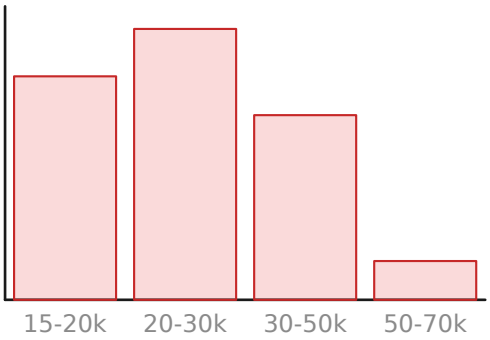
$$360 - 150 - 120 = 90°, \qquad \frac{90}{360} \times 120 = 30 \text{ people}$$

what's left of 360°

Case 1: the HMRC wage brackets

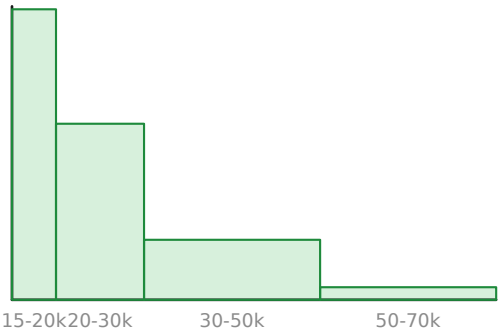
Lecture 4 shows how many UK taxpayers earn in each wage bracket and asks: **"Something is wrong with my choice of graph here. What is it?"** Here are the middle brackets:

SUSPECT: equal-width bars



20-30k looks like the winner

HONEST: width = bracket size



The brackets are **different widths**: £15k to £20k is only £5000 wide, but £30k to £50k is £20,000 wide. A wide bracket collects more people just by being wide. The fix (a proper **histogram**) makes the bar **width** the bracket width and the **area** the number of people, so the height becomes **people per £1000**:

$$\frac{4.95 \text{ million}}{5} = 0.99$$

million per £1000, for £15-20k: the real crowd

$$\frac{6.00 \text{ million}}{10} = 0.60$$

million per £1000, for £20-30k: thinner!

Verdict: the busiest **£1000-slice** is around £15k-£20k, not £20k-£30k. This is why the lecture's mode lands at about £15,000 and the median at £16,400 (Chapter 24).

Histograms with unequal widths: the rule

RULE: frequency density

In a histogram the **area** of a bar is the frequency. The bar's **height** is the **frequency density**, and the vertical axis is labelled "frequency density".

frequency density $= \dfrac{\text{frequency}}{\text{class width}}$, where class width = upper boundary − lower boundary.

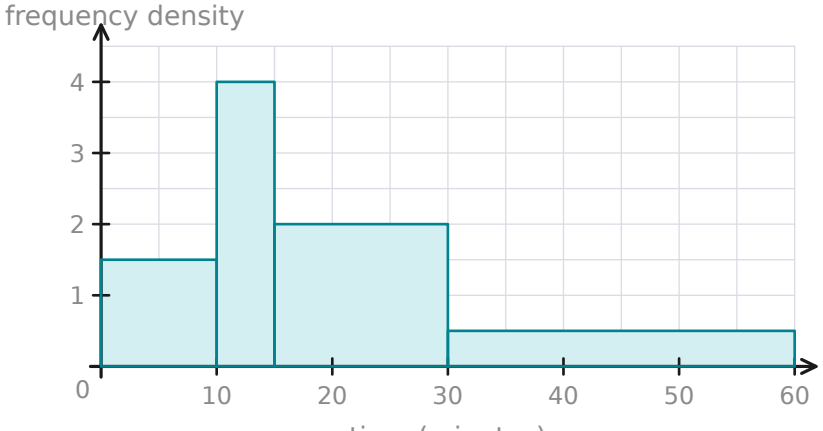
Going back: frequency = frequency density × class width (the area of the bar).

If every class has the same width you may plot plain frequency instead: the shape is the same.

PROCEDURE: reading frequencies off a histogram

1. Read each bar's class boundaries off the horizontal axis and work out its width.
2. Read the bar's height (the frequency density) off the vertical axis.
3. Multiply: frequency = density × width. For "how many between ..." add the bars (or the parts of bars) inside that range.

Example: the commuting times of a group of people. How many took each time, and how many people were asked?



tallest bar (10-15) is not the biggest group: it is only 5 minutes wide

0 to 10: $1.5 \times 10 = 15$ ← *width 10*

10 to 15: $4 \times 5 = 20$ ← *width 5*

15 to 30: $2 \times 15 = 30$ ← *width 15*

30 to 60: $0.5 \times 30 = 15$ ← *width 30*

$$15 + 20 + 30 + 15 = 80 \text{ people}$$ ← *total = sum of the areas*

And forwards again: if 12 more people took 60 to 80 minutes, that bar's height would be $12 \div 20 = 0.6$.

Stem and leaf, one leaf at a time

RULE: stem and leaf diagram

Leaf = the last digit; **stem** = everything in front of it. 34 is stem 3, leaf 4; 5 is stem 0, leaf 5.

Write every stem once, in order, down the left of a vertical line, **including empty ones**. Put each leaf beside its stem, smallest next to the line.

Always give a **key**: "2 | 5 means 25" (or 2.5, or 250: without the key nobody knows).

Computer versions (Lecture 5's Minitab) may split each stem into two rows (leaves 0-4, then 5-9) and add a left-hand column counting the values from each end, with the row holding the median shown in brackets.

Lecture 4's data set, 17 values already in order:

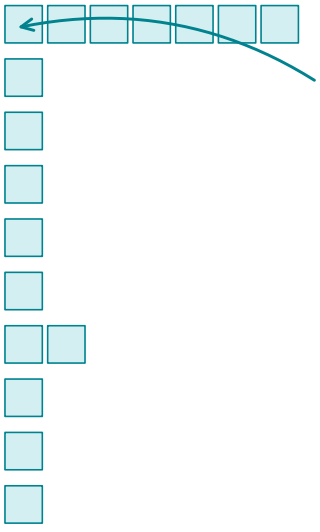
1 2 3 5 5 6 8 18 25 34 47 56 60 69 71 88 94

Split every number into a stem (the tens digit) and a leaf (the units digit). 34 splits into stem 3, leaf 4. Single figures like 5 have stem 0.

34 → 3 | 4 5 → 0 | 5 94 → 9 | 4

0	1 2 3 5 5 6 8
1	8
2	5
3	4
4	7
5	6
6	0 9
7	1
8	8
9	4

key: 2 | 5 means 25



tip it on its side and it's a bar chart of the tens:

7 values under 10, then a long thin tail up to 94

You keep **every original number** (unlike a bar chart) and see the **shape** at a glance. Always give a **key**, or nobody knows if 2 | 5 is 25 or 2.5.

Case 2: the lecturer in the room

Lecture 5: "Consider the distribution of ages in this room." Twelve students and one lecturer:



An **outlier** is a value far from the rest. It drags the average: the mean age is **22.3** with the lecturer and **19.8** without. But the lecturer is **real**: they really are in the room!

RULE: outliers (Lecture 5)

Remove a point only if you think something is **wrong** with it (a typo like age 190, a broken sensor). "Unusual" is not the same as "wrong".

If you do remove one, **always say that you removed it and why**.

Describing a scatter plot: correlation words

RULE: describing correlation

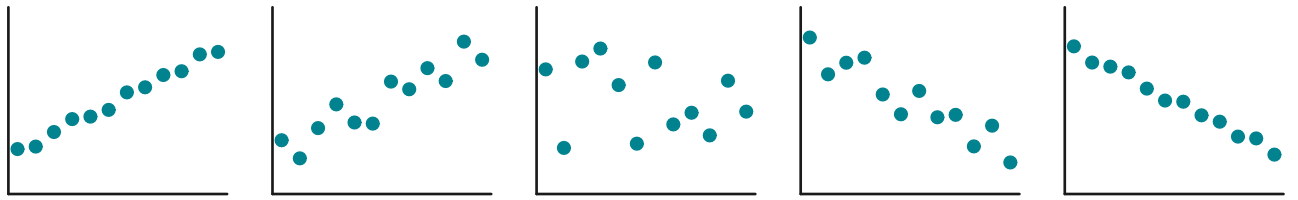
Positive correlation: as one measurement goes up, the other tends to go up (the cloud slopes uphill).

Negative correlation: as one goes up, the other tends to go down (the cloud slopes downhill).

No correlation: no uphill or downhill trend, just a shapeless cloud.

Strong: the points sit close to a straight line. **Weak**: a loose cloud with only a rough trend. Always say both, e.g. "strong negative".

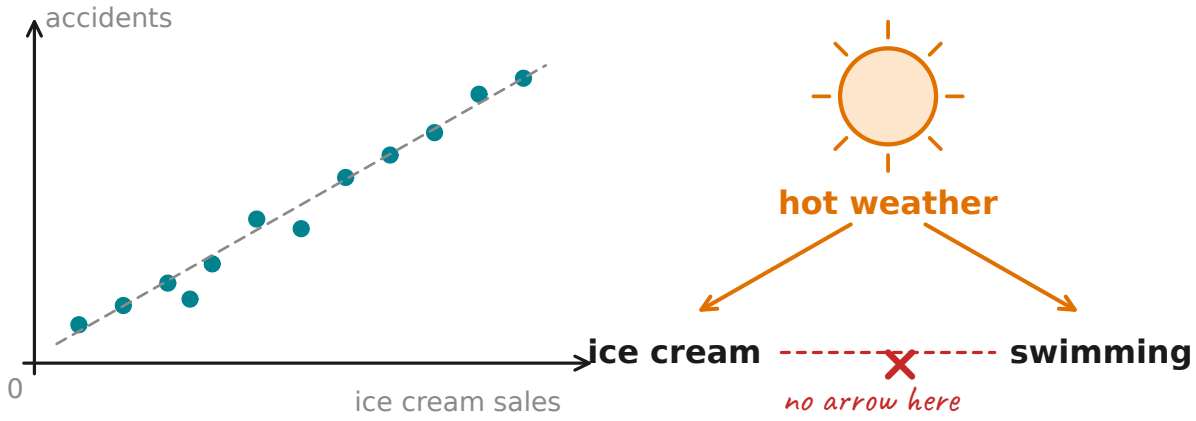
Correlation describes a pattern; it does **not** show that one thing causes the other (next case).



strong positive | weak positive | none | weak negative | strong negative

Case 3: does ice cream cause drowning?

A town records ice cream sales and swimming accidents each month. The scatter plot shows a clear pattern: more ice cream, more accidents. Lock up the ice cream van?



The two **go together** (**correlation**), but neither **makes** the other happen (**causation**). A hidden third thing, hot weather, pushes both up. Lecture 5: **correlation does not equal causation!**

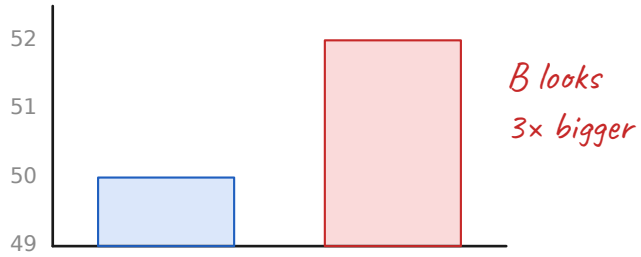
Direction trap: even when there is a real cause, which way does it go? Towns with more police have more crime. Do police cause crime, or does crime bring police?

Joining the dots: only connect the points with a line when there's a reason, like measurements taken one after another in **time**. Here, a line through the months in sales order would be meaningless; a dashed **trend line** is fine.

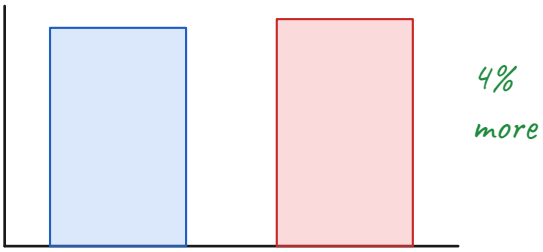
Case 4: crooked graphs and their honest twins

Lecture 5: bad graphs hide or twist information: missing titles or labels, poor use of colour, and a **poor choice of scale**. Each crime on the left, the fix on the right.

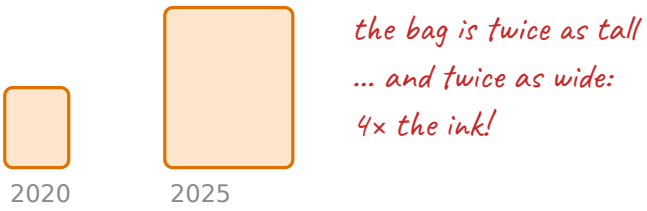
CRIME: axis starts at 49



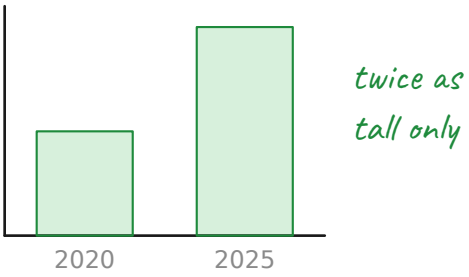
HONEST: axis from 0



CRIME: picture grows both ways



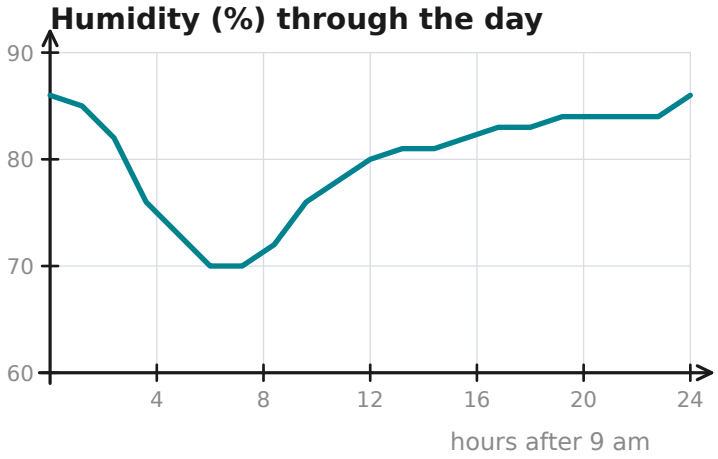
HONEST: plain bars



The checklist of crimes: no title, no axis labels or units, no source, a y-axis that doesn't start at 0 (for bars), uneven scales, 3D effects, pictures that grow in area, and "infographics" (Lecture 5: *very modern, very useful, very easy to lie with!*).

Reading any graph in three passes

Lecture 5's method: **start with how it's built, move on to what it says, finish by being critical.** Try it on this graph:



1 FORM

Title? Yes. Axes? % and hours, but where is the source? Why start at 60?

2 MESSAGE

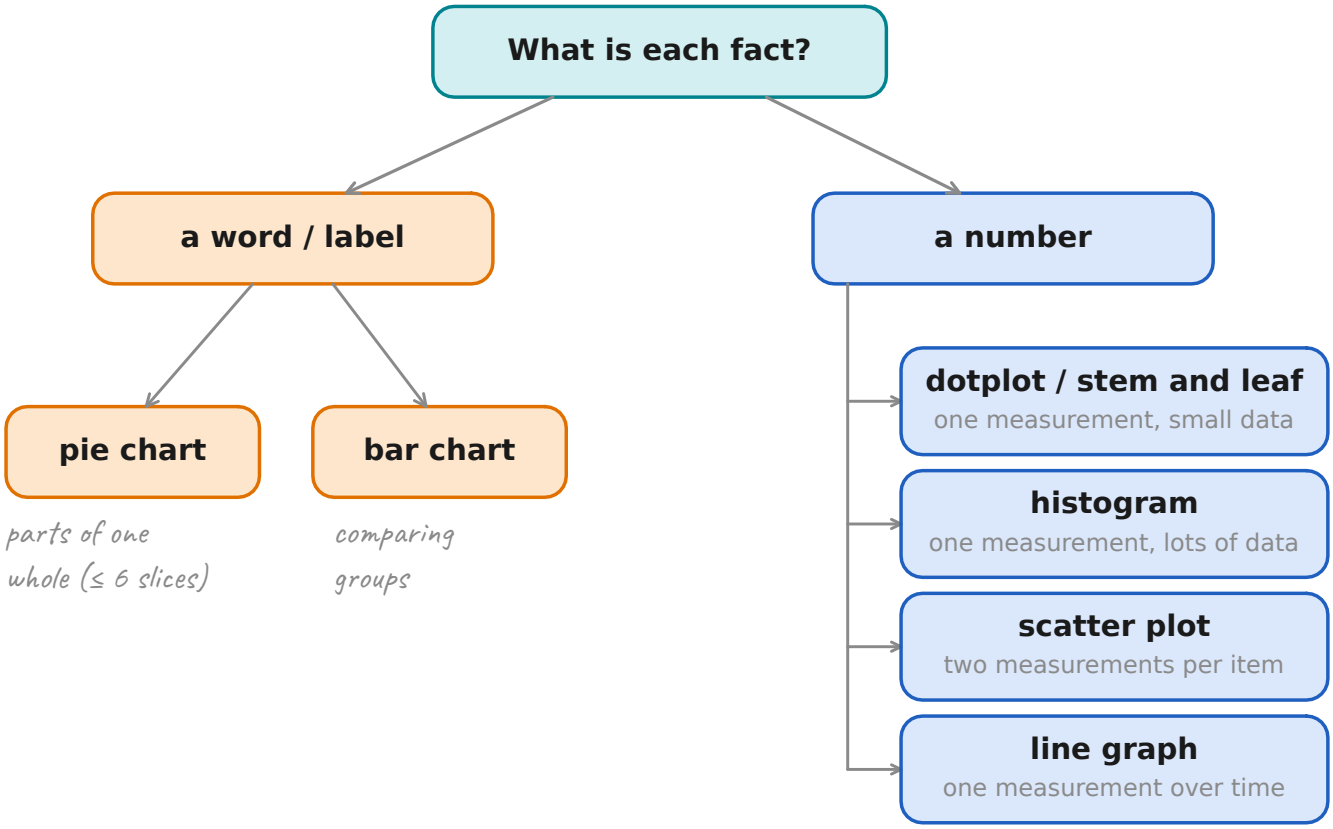
Driest around 3–4 pm (70%), damp again overnight. Implication: sun warms the air and dries it.

3 CRITICISM

One day only, one place. Missing source. Axis from 60 exaggerates the dip.

Form: title and footnotes, axis labels and scale, data source, colours and symbols. **Message:** meaning and implications. **Criticism:** missing data and ease of understanding. (The data are the weather table from the lecture slides.)

Which chart? A flowchart



And when is a table better than any chart? When the reader needs **exact values** to look up, not a shape. Lecture 5's weather forecast is a table because you want to know the temperature at 3 pm, not admire a curve:

time	9:00	12:00	15:00	18:00	21:00	0:00	3:00
temp °C	10	12	12	11	10	10	9
wind	N	NNE	NNE	NNE	NNE	N	N
humidity	86%	76%	70%	78%	81%	83%	84%

Rule of thumb: tables for **looking up**, charts for **seeing patterns**. A table needs the same honesty as a graph: a title, units in the headings, and a source.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Pie chart angle	$\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$ all the angles add up to 360°
Pie chart frequency	$\text{frequency} = \frac{\text{angle}}{360°} \times \text{total}$
Frequency density	$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$ the height of a histogram bar; label the axis "frequency density"
Frequency from a histogram	$\text{frequency} = \text{frequency density} \times \text{class width}$ frequency = area of the bar
Bar chart vs histogram	bar chart: categories, gaps between bars. Histogram: numbers in ranges, bars touch, area = frequency
Stem and leaf	leaf = last digit, stem = the rest; leaves in order; always give a key
Correlation	positive (uphill), negative (downhill) or none; strong (close to a line) or weak
Correlation vs causation	a pattern between two things does not show that one causes the other
Outliers	remove a point only if it is wrong (not just unusual), and say that you did and why
Honest graphs	title, labelled axes with units, a source; bar charts start at 0; no pictures that grow in area
Reading a graph	form (how it is built), message (what it says), criticism (what is missing or misleading)

Case files for you

Answers in grey.

Which chart?

1. Share of the budget spent on rent, food, travel, fun
2. Reaction times of 200 people
3. Hours of revision vs exam mark for 30 students
4. Temperature at noon every day in March

1. pie 2. histogram 3. scatter plot 4. line graph

Build it

5. Draw a stem and leaf for 12, 15, 21, 22, 22, 27, 30, 34, 41 (with a key)
6. A pie chart shows 90 people. How many degrees for 25 of them?

5. 1 | 2 5; 2 | 1 2 2 7; 3 | 0 4; 4 | 1; key 2 | 1 means 21 6. 25/90 × 360 = 100°

Detective

7. "Students who sit at the front get better marks, so sitting at the front makes you clever." What's wrong?
8. A bar chart's y-axis runs from 95 to 100. Two bars read 96 and 99. What false impression, and the fix?
9. Data: 4, 5, 5, 6, 7, 480. Should you delete 480?

7. correlation ≠ causation (keen students choose the front) 8. 99 looks 4× bigger than 96; start at 0 9. only if it's an error; check it, and say so if you remove it

Find the unknown

10. A pie chart shows 250 people. One slice is 72°. How many people is that?
11. A 45° slice stands for 30 people. How many people does the whole pie show?
12. A histogram bar for 20-30 cm has frequency density 3.5. What is its frequency?
13. A class 5-8 kg holds 24 parcels. How tall is its histogram bar?
14. Bars: 0-10 at density 2, 10-15 at density 6, 15-30 at density 1. How many values altogether?
15. A pie of 120 people has slices of 150°, 120° and x°. Find x and that slice's frequency.

10. 72/360 × 250 = 50 11. 45° is 1/8 of 360°, so 30 × 8 = 240 12. 3.5 × 10 = 35 (frequency = density × width) 13. 24 ÷ 3 = 8 (density = frequency ÷ width) 14. 2 × 10 + 6 × 5 + 1 × 15 = 20 + 30 + 15 = 65 15. x = 360 − 150 − 120 = 90°; 90/360 × 120 = 30